

BRAC UNIVERSITY
Department of Computer Science and Engineering
Semester: Summer2024
Section-9, 10, 12

Long Multiplication

if (last bit of multiplier = 1)

1. Product = multiplicand + product
2. 1 bit left shift of multiplicand
3. 1 bit right shift of the multiplier

if (last bit of multiplier = 0)

1. 1 bit left shift of multiplicand
2. 1 bit right shift of the multiplier

Number Of Iteration = Number of bits in multiplier

8 * 9 = 72 in 4 bit architecture

	Multiplicand 0000 1000	Multiplier 1001	Product 0000 0000
1	0000 1000	1001	0000 1000
	0001 0000	1001	0000 1000
	0001 0000	0100	0000 1000
2	0010 0000	0100	0000 1000
	0010 0000	0010	0000 1000
3	0100 0000	0010	0000 1000
	0100 0000	0001	0000 1000
4	0100 0000	0001	0100 1000
	1000 0000	0001	0100 1000
	1000 0000	0000	0100 1000

Optimized Multiplication

if (last bit of multiplier = 1)

1. MSB of Product = multiplicand + product MSB half bits
2. 1 bit right shift product

if (last bit of multiplier = 0)

1 bit right shift product

Number Of Iteration = Number of bits in multiplier

8 * 9 = 72 in 4 bit architecture

	Multiplicand 1000	Product 0000 1001
1	1000	1000 1001
		0100 0100
2	1000	0010 0010
3	1000	0001 0001
4	1000	1001 0001
		0100 1000

Converting into Floating Point Representation

Question:

Convert 70.8863 to 32 bit IEEE 754
Floating Point Representation.

Step 1: Decimal to binary conversion.

$$70 = 1000110$$
$$70.8863 = 1000110.111000101$$

$$\begin{array}{r} 0.8863 \\ \times 2 \\ \hline 1.7726 \\ \times 2 \\ \hline 1.5452 \\ \times 2 \\ \hline 1.0904 \\ \times 2 \\ \hline 0.1808 \\ \times 2 \\ \hline 0.3616 \\ \times 2 \\ \hline 0.7232 \\ \times 2 \\ \hline 1.4464 \\ \times 2 \\ \hline 0.8928 \\ \times 2 \\ \hline 1.7856 \end{array}$$

$$0.8863 = 0.111000101$$

Step-02 : Normalize the binary number

1000110 · 111000101

↪ $1.000110111000101 \times 2^6$

NOTE: If you were told in the question that you have to calculate exact 20 bits for the fractional part, we had to calculate for 5 bits by the multiplication.

Here we get exact 15 bits for the fraction out of 23 bits.

Step-03: Calculate bias and find out biased exponent.

for 8 bit ~~for~~ exponent field

we get bias value = $2^{n-1} - 1$

$$= 2^{8-1} - 1 = 127$$

$$\therefore \text{Biased exponent} = 6 + 127 = 133$$

$$133 = 10000101$$

Step - 04: Find out fraction and sign bit.

Sign bit = 0

fraction = 0001 1011 000101 00000000

Step-05: Represent

sign bit	Exponent	fraction
1 bit	8 bit	23 bit

0	1 00000 101	0001 1011 000101 00000000
sign bit	exponent	fraction

Floating Point to Decimal Conversion

Question:

0x F2400120

This number is in single precision floating point representation. What is the decimal value?

Step-01: Hexadecimal $\rightarrow$ Binary

F2400120

$\hookrightarrow$ 1111 0010 0100 0000 0000 0001 0010 0000

Step-02: Organize binary number into format.

<u>1</u>	<u>1110 0100</u>	<u>1 000000 0000 0001 0010 0000</u>
sign bit	biased exponent	fraction

Step-03: find out exponent and fraction

$$\text{Biased exponent} = 11100100 = 228$$

$$\text{bias} = 2^{n-1} - 1 = 127$$

$$\therefore \text{exponent} = 228 - 127 = 101$$

$$\text{fraction} = 100000000000000100100000$$

$$= (1 \times 2^{-1}) + (1 \times 2^{-15}) + (1 \times 2^{-18})$$

$$= 0.5000343323$$

Step-04: Use formula:

$$\text{Decimal value} = (-1)^s \times (1 + \text{fraction}) \times 2^{\text{exponent}}$$

$$= (-1)^1 \times (1 + 0.5000343323) \times 2^{101}$$

$$= -1.5000343323 \times 2^{101}$$

$$= -3.803038843 \times 10^{30}$$

Convert the number 20.7895 into a floating point format where total bit length would be 20 and 6 bits will be allocated for the biased exponent field